

TOTAL NO. OF QUESTIONS: 5]

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BE-102(GS)

B. E. (First/Second Semester)

EXAMINATION, June, 2012

(Grading System)

(Common for all Branches)

ENGINEERING MATHEMATICS-I [BE – 102(GS)]

Time: Three Hours Maximum Marks: 70

Minimum Pass Marks: 22 (D Grade)

Note: Attempt all questions. All questions carry equal marks.

1. (a) Prove that:

$$\tan^{-1}(x+h) = \tan^{-1}x + h \sin z \frac{\sin z}{1} - \frac{(h \sin z)^2}{2} \sin 2z + \dots$$

where $z = \cot^{-1}x$.

(b) If $u = x\varphi(y/x) + \psi(y/x)$, then prove that:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$$

Or

(a) What error in the common logarithm of a number will be produced by an error of 1% in the number?

(b) Find the maxima and minima of the following function:

$$\sin x + \sin y + \sin(x+y) \text{ in } \left[0 \leq x \leq \frac{\pi}{2}, 0 \leq y \leq \frac{\pi}{2}\right].$$

2. (a) Find ab-initio the value of the integral:

$$\int_0^{\pi/2} \sin x \, dx$$

(b) Evaluate:

$$\int_0^\infty \frac{8(1-x^6)}{(1+x^2)^4} dx$$

Or

(a) Evaluate:

$$\lim_{n \rightarrow \infty} \left(\frac{1^n}{n^n} \right)^{1/n}$$

(b) Change the order of integration:

$$\int_0^4 \int_{x/4}^x dx \, dy$$

Hence evaluate it.

3. (a) Solve:

$$y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0$$

(b) Solve:

$$\frac{d^2y}{dx^2} + 4y = e^x + \sin 2x$$

Or

(a) Solve:

$$p(py) = x(x + y)$$

where $p = \frac{dy}{dx}$.

(b) Solve:

$$\begin{aligned}\frac{dx}{dt} + y &= \sin t \\ \frac{dy}{dt} + x &= \cos t\end{aligned}$$

Given that $x = 2$ and $y = 2$, when $t = 0$.

4. (a) Find the rank of the matrix:

$$A = \begin{bmatrix} 1 & 4 & 3 & 6 & 1 \\ 0 & 2 & 1 & 4 & 4 \\ 0 & 0 & 1 & 3 & 7 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{5 \times 5}$$

by defining it in Echelon form.

(b) Find the eigen values and eigen vectors of the matrix:

$$A = \begin{bmatrix} 3 & -4 & 4 \\ 1 & -2 & 4 \\ 1 & -1 & 3 \end{bmatrix}$$

Or

(a) Find the values of k such that the system of equations:

$$x + ky + 3z = 0$$

$$4x + 3y + kz = 0$$

$$2x + y + 2z = 0$$

has non-trivial solution.

(b) Verify the Cayley-Hamilton theorem for the matrix:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

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5. (a) Define the following terms for a graph:

(i) Subgraph

(ii) Degree of a vertex

(iii) Composition and De-composition

(iv) Rooted tree

(b) Define fuzzy logic and its applications in science and engineering.

Or

(a) "Prepare a truth table to get the negation of the statement: "Sita is dull and careless.""

(b) Prove that:

$$a \cdot b + b \cdot c + c \cdot a = (a + b) \cdot (b + c) \cdot (c + a) \forall a, b, c, \in B$$